

Tips for Improving Impact of your Scientific Communication

(Developed for writing but applies to your Elevator Pitch too)

Remember your Audience is you scientifically literate peers (the outside world).

Your aim is to help your peers see something that might deepen their understanding of the topic beyond just what they would get from reading the background for themselves.

Strategies for deeper analysis [Impact!]

- **Before/after** - communicate how your analysis (will) change/enrich the way we think about topic
- **Connect-the-dots** - Make connections between assumptions, what your analysis showed (or will show), and implications

Strategies for improved flow

- **Old before New** - order of information is key! (background before new work):

Example of student writing:

XX was one of the first freshwater fish of the Carboniferous period. Its digestive system anatomy was modeled after its closet living relative, YY, due to its similar diet and living conditions. XX preyed on prehistoric amphibians, bony fish, and sharks. The YY is a bony fish that was used as a model, since it's subclass was present during the Carboniferous and likely fell prey to XX.

Same content, organized **Old** before **New**:

XX was one of the first freshwater fish of the Carboniferous period. It is believed to have preyed on prehistoric amphibians, bony fish, and sharks. To model its diet, we used the nutritional contents of the modern bony fish YY as members of its subclass were present during the Carboniferous and likely fell prey to XX.

- **Topics as subjects of sentences** - assist audience with prioritizing information

Example:

The Oligocene period (33.9-23 MYA) was an epoch that experienced a decrease in temperature, which could have affected XX's diet. XX was considered a forest-dwelling browser due to its low-crowned teeth. However, teeth in YY evolution and climate change could show that it had an herbivorous diet consistent with browsing and grazing habits.

Topics as subjects and Connect-the-dots:

Cooling of the climate during the Oligocene period (33.9-23 MYA) could have affected XX's diet. Prior to this time, XX was considered a forest-dwelling browser due to its low-crowned teeth. Conversion of forests to open grasslands could explain the observed evolutionary trend toward grinding molars in this group. Therefore, climate change could have promoted an herbivorous diet consistent with both browsing and grazing habits.